Elhama Forest

Author : MOUGARI Aymen

Difficulty Level : Hard

Time Limit : 0.23 s

Description

In the mystical region of Elhama Forest, the ancient trees are connected by deep roots forming a powerful magical network — a tree of N nodes rooted at node 1. Each tree holds an Ethereum core, a special energy that sustains the forest's life force.

Every ith node has a value ai representing its Ethereum core.

Halim, the guardian of the forest, can cast two kinds of spells:

- Infuse Spell (1 u val): Given a node u and a value **val**, update the entire subtree rooted at u, replacing all Ethereum core values in that subtree with **val — excluding** the root node **u** itself.

- Inspect Spell (2 u): Given a node **u**, determine whether the sum of all Ethereum core values in the subtree rooted at **u** (including **u** itself) is a prime number.

(The sum of subtree nodes including the subtree root u => is a prime number).

Input

- The first line contains one integer n ($1 \le n \le 10^3$) — the number of nodes in the tree.

- The second line contains n integers a1, a2, ..., a (ai may be a very big number) — the initial Ethereum core value of each node.

- The next n - 1 lines each contain two integers u and v ($1 \le u, v \le n, u \ne v$), representing an edge connecting node u with node v.

- The next line contains a single integer q ($1 \le q \le 500$) — the number of queries.

- Each of the following q lines describes a query:

- 1 u val — Infuse Spell: update all Ethereum cores in the subtree rooted at u to val, excluding node u itself.

- 2 u — Inspect Spell: determine whether the sum of the Ethereum cores in the subtree rooted at u (including u) is a prime number.

Output

For each query of the second type, print “**YES**” if the sum is a prime number, otherwise print “**NO**”.

Examples

Input	Output
5 1 9 3 4 5 1 2 1 3 2 4 2 5 3 1 2 10 2 2 2 1	YES NO
5 1 0 1 43 32 1 2 2 3 3 4 4 5 4 1 3 295779363 2 3 1 1 8935637299718154180477378410603085824 2 1	YES NO